

DIFFERENTIALS, DUAL ISOGENIES, AND THE TATE MODULE: SOLUTIONS

SANTIAGO ARANGO-PÍÑEROS

Here are three computations by hand, on the invariant differential, the dual isogeny and the Tate module. Each compares the multiplicative group $\mathbb{G}_m$ with the elliptic curve

$$E: y^2 = x^3 - x, \quad \omega_E := \frac{dx}{2y}.$$

Throughout, k is a field with $\text{char } k \neq 2$, $\bar{k}$ an algebraic closure and $k^{\text{sep}} \subseteq \bar{k}$ the separable closure. Here $\mathbb{G}_m = \mathbb{P}_k^1 \setminus \{0, \infty\}$ has coordinate t and invariant differential $\omega_{\mathbb{G}_m} := dt/t$. Each problem is preceded by the definitions and results it uses.

Reading. [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Proposition 3.1.

A formula. We use the following duplication formula on E , derived in Problem 1.1(5): if $x \in k(E)$ is the x -coordinate and $[2]: E \rightarrow E$ is multiplication by 2, then

$$(0.1) \quad [2]^*x := x \circ [2] = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

1. INVARIANT DIFFERENTIALS

Problem 1.1 (Poles and pullbacks).

(1) Rewrite $\omega_{\mathbb{G}_m}$ in the coordinate $s := 1/t$ near ∞ , and compute $\text{div}(\omega_{\mathbb{G}_m})$ on $\mathbb{P}_k^1$. Compute $[m]^*\omega_{\mathbb{G}_m}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.

SOLUTION. $\omega_{\mathbb{G}_m} = -ds/s$, $\text{div}(\omega_{\mathbb{G}_m}) = -(\mathbf{0}) - (\infty)$ and $[m]^*\omega_{\mathbb{G}_m} = m\omega_{\mathbb{G}_m}$. $t = 1/s$ gives $dt = -ds/s^2$, so $\omega_{\mathbb{G}_m} = -ds/s$. Hence $\omega_{\mathbb{G}_m}$ has simple poles at 0 and ∞ , with residues 1 and -1 , and no other zeros or poles: $\text{div}(\omega_{\mathbb{G}_m}) = -(\mathbf{0}) - (\infty)$. Also $[m]^*\omega_{\mathbb{G}_m} = d(t^m)/t^m = m\omega_{\mathbb{G}_m}$.

(2) Show that $\omega_E = dy/(3x^2 - 1)$, and that y and $3x^2 - 1$ have no common zero on E . Deduce that ω_E has no zeros or poles on $E \setminus \{O\}$.

SOLUTION. Differentiating $y^2 = x^3 - x$ gives $2y dy = (3x^2 - 1) dx$. If $y = 0$, then $x \in \{0, \pm 1\}$ and $3x^2 - 1 \in \{-1, 2\}$, which is non-zero. On the open set $y \neq 0$ the differential dx generates $\Omega_{E/k}$, and $\omega_E = dx/(2y)$ is a unit times it. On $3x^2 - 1 \neq 0$, likewise with dy and $dy/(3x^2 - 1)$. These open sets cover $E \setminus \{O\}$.

(3) Using $\text{ord}_O x = -2$ and $\text{ord}_O y = -3$, show that $\text{ord}_O \omega_E = 0$.

SOLUTION. With a uniformizer w at O , write $x = w^{-2}u$ with $u(O) \neq 0$. Then

$$dx = (-2w^{-3}u + w^{-2}u') dw$$

has order -3 , because $2 \neq 0$ in k . So $\text{ord}_O \omega_E = -3 - (-3) = 0$.

(4) Compute $[-1]^*\omega_E$ from $[-1](x, y) = (x, -y)$, and compare with $[-1]^*\omega_{\mathbb{G}_m}$.

SOLUTION. $[-1]^*\omega_E = -\omega_E$, and $[-1]^*\omega_{\mathbb{G}_m} = -\omega_{\mathbb{G}_m}$. $[-1]^*\omega_E = dx/(-2y) = -\omega_E$. On $\mathbb{G}_m$, $[-1](t) = 1/t$ and $[-1]^*\omega_{\mathbb{G}_m} = -\omega_{\mathbb{G}_m}$ as well.

(5) We derive (0.1) by hand. Here $x, y \in k(E)$ are functions. For a point $P \in E(\bar{k})$ write $x_P := x(P)$ and $y_P := y(P)$ for its coordinates.

- (a) Let $P \in E(\bar{k})$ with $y_P \neq 0$. Show that the tangent line to E at P has slope $\lambda_P = (3x_P^2 - 1)/(2y_P)$, and use that its third point of intersection with E is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.

SOLUTION. In the affine plane with coordinates x, y , differentiating $y^2 = x^3 - x$ at P gives $2y_P dy = (3x_P^2 - 1) dx$, so the tangent line at P is

$$y = \lambda_P x + \nu_P, \quad \lambda_P = \frac{3x_P^2 - 1}{2y_P}, \quad \nu_P = y_P - \lambda_P x_P.$$

Substituting into $y^2 = x^3 - x$ gives

$$x^3 - \lambda_P^2 x^2 - (1 + 2\lambda_P \nu_P)x - \nu_P^2 = 0,$$

whose roots are x_P , twice because the line is tangent, and $x(-[2]P)$. Comparing coefficients of x^2 gives $x(-[2]P) = \lambda_P^2 - 2x_P$, and $x(-[2]P) = x([2]P)$ since $[-1](x, y) = (x, -y)$.

- (b) Let $\lambda := (3x^2 - 1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.

SOLUTION. $[2]^*x = (x^2 + 1)^2/(4(x^3 - x))$. By definition of the pullback, $([2]^*x)(P) = x([2]P)$, and $(\lambda^2 - 2x)(P) = \lambda_P^2 - 2x_P$. So by (5)(a) the function $[2]^*x - (\lambda^2 - 2x) \in k(E)$ vanishes at every $P \in E(\bar{k})$ with $y_P \neq 0$, that is, at every point of $E(\bar{k})$ outside the finite set $E[2](\bar{k})$. A non-zero rational function has only finitely many zeros, so $[2]^*x = \lambda^2 - 2x$ in $k(E)$. Finally, in $k(E)$ we have $4y^2 = 4(x^3 - x)$, so

$$[2]^*x = \lambda^2 - 2x = \frac{(3x^2 - 1)^2 - 8x(x^3 - x)}{4(x^3 - x)} = \frac{x^4 + 2x^2 + 1}{4(x^3 - x)} = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

Remark. This is the precise meaning of the shorthand “ $x([2]P) = \lambda^2 - 2x$ for $P = (x, y)$ ”.

- (6) Solve this part in **Magma**, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function λ of (5), check (0.1), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \quad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2 + 1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

SOLUTION. **Both signs are +.** The identities have coefficients in $\mathbb{Z}[1/2]$, so it suffices to check them over $\mathbb{Q}$. By (5), $[2]^*x = \lambda^2 - 2x$ with $\lambda = (3x^2 - 1)/(2y)$, and the same argument applied to the y -coordinate of the third point of intersection gives $[2]^*y = -(y + \lambda([2]^*x - x))$. Since $\omega_E = dx/(2y)$, we have

$$\frac{[2]^*\omega_E}{\omega_E} = \frac{d([2]^*x)/dx}{2[2]^*y} \cdot 2y.$$

```
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>; // q(E)
lam := (3*x^2 - 1)/(2*y); // tangent slope
```

```

x2 := lam^2 - 2*x; y2 := -(y + lam*(x2 - x)); // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x)); // true
[ Factorization(Numerator(Kx!(x2 - e))) : e in [0, 1, -1] ];
// [ [ <x^2 + 1, 2> ], [ <x^2 - 2*x - 1, 2> ], [ <x^2 + 2*x - 1, 2> ] ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3); // true
(F!Derivative(Kx!x2))/(2*y2) * 2*y; // 2

```

The denominator of $[2]^*x - e$ is that of $[2]^*x$, namely $4(x^3 - x)$, and the factorizations show that its numerator is $g_e(x)^2$. So $[2]^*y = (x^2 + 1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = 2\omega_E$, as [Sil09, Corollary III.5.3] predicts.

Remark. Alternatively, **Magma**'s elliptic curve functionality provides $[2]$ as an isogeny, and pulls back both functions and differentials along it, so no group law is typed in.

```

E := EllipticCurve([-1,0]); // y^2 = x^3 - x
m2 := MultiplicationByMMap(E, 2); // [2] : E -> E
KE<x,y> := FunctionField(E); // Q(E)
x2 := Pullback(m2, x); y2 := Pullback(m2, y); // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x)); // true
g := [x^2 + 1, x^2 - 2*x - 1, x^2 + 2*x - 1]; // g_0, g_1, g_{-1}
[ x2 - [0, 1, -1][i] eq g[i]^2/(4*(x^3 - x)) : i in [1..3] ];
// [ true, true, true ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3); // true
omega := Differential(x)/(2*y); // omega_E
Pullback(m2, omega)/omega; // 2

```

2. DUAL ISOGENIES

Consider the elliptic curves

$$E_1: y_1^2 = x_1^3 - x_1, \quad E_2: y_2^2 = x_2^3 + 4x_2, \quad E_3: y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i := dx_i/(2y_i)$ for the invariant differential of E_i . Consider the maps

$$\begin{aligned} \varphi: E_1 &\rightarrow E_2, \quad (x_1, y_1) \mapsto \left(\frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2 + 1)}{x_1^2} \right), \\ \psi: E_2 &\rightarrow E_3, \quad (x_2, y_2) \mapsto \left(\frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2 - 4)}{x_2^2} \right), \\ \iota: E_3 &\rightarrow E_1, \quad (x_3, y_3) \mapsto \left(\frac{x_3}{4}, \frac{y_3}{8} \right). \end{aligned}$$

The maps φ and ψ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies. Also ι is an isomorphism. That example computes the kernels and the dual, but not the constants c_α of Definition 2.1: that is part (2) of Problem 2.2.

Definition 2.1 (The constant c_α). For an isogeny $\alpha: E_i \rightarrow E_j$, define c_α by $\alpha^*\omega_j = c_\alpha\omega_i$.

Problem 2.2 (A 2-isogeny and its dual).

(1) Check that φ lands on E_2 . Show that $\ker \varphi = \{O_1, (0, 0)\}$, that $\varphi(\pm 1, 0) = (0, 0)$, and that $\deg \varphi = 2$.

SOLUTION. Since $y_1^2 = x_1(x_1^2 - 1)$, the functions $\varphi^*x_2 = (x_1^2 - 1)/x_1$ and $\varphi^*y_2 = y_1(x_1^2 + 1)/x_1^2$ satisfy

$$(\varphi^*y_2)^2 = \frac{x_1(x_1^2 - 1)(x_1^2 + 1)^2}{x_1^4}, \quad (\varphi^*x_2)^3 + 4\varphi^*x_2 = \frac{x_1^2 - 1}{x_1} \cdot \frac{(x_1^2 + 1)^2}{x_1^2},$$

which agree. The function $\varphi^*x_2 = x_1 - 1/x_1$ has poles only at $(0, 0)$ and at O_1 , so $\ker \varphi = \{O_1, (0, 0)\}$. And at $(\pm 1, 0)$ both coordinates of φ vanish. Next,

$$y_1 = \frac{x_1^2 \varphi^*y_2}{x_1^2 + 1}, \quad x_1^2 - (\varphi^*x_2)x_1 - 1 = 0,$$

so $k(E_1) = \varphi^*k(E_2)(x_1)$ has degree at most 2 over $\varphi^*k(E_2)$. Since $\#\ker \varphi = \deg_s \varphi \leq \deg \varphi$, we get $\deg \varphi = 2$.

(2) Compute c_φ .

SOLUTION. $c_\varphi = 1$.

$$\varphi^*dx_2 = \left(1 + \frac{1}{x_1^2}\right) dx_1 = \frac{(x_1^2 + 1) dx_1}{x_1^2}, \quad 2\varphi^*y_2 = \frac{2y_1(x_1^2 + 1)}{x_1^2},$$

so $\varphi^*\omega_2 = \omega_1$ and $c_\varphi = 1$.

Remark. This is why the second coordinate of φ has the sign it does: the opposite sign would give $c_\varphi = -1$.

(3) Check that ψ lands on E_3 and that ι lands on E_1 , and compute c_ψ and c_ι .

SOLUTION. $c_\psi = 1$ and $c_\iota = 2$. φ and ψ are the same construction for $y^2 = x^3 + bx$, with $b = -1$ and $b = 4$ respectively, and target $y^2 = x^3 - 4bx$. The computation of (1) applies verbatim, and that of (2) gives $c_\psi = 1$. Also $(y_3/8)^2 = (x_3^3 - 16x_3)/64 = (x_3/4)^3 - x_3/4$, so ι lands on E_1 , and

$$\iota^*\omega_1 = \frac{dx_3/4}{2y_3/8} = 2\omega_3,$$

so $c_\iota = 2$.

(4) Show that $(\iota \circ \psi \circ \varphi)^*x_1 = [2]^*x_1$, so that $\iota \circ \psi \circ \varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat{\varphi} = \iota \circ \psi$, that $c_\varphi c_{\hat{\varphi}} = \deg \varphi$, and that $\ker \hat{\varphi} = \varphi(E_1[2](\bar{k}))$.

SOLUTION. **The sign is +: $\iota \circ \psi \circ \varphi = [2]$, and $\hat{\varphi} = \iota \circ \psi$.** We have $\psi^*x_3 = y_2^2/x_2^2 = x_2 + 4/x_2$ and $\iota^*x_1 = x_3/4$, so

$$(\iota \circ \psi \circ \varphi)^*x_1 = \frac{1}{4} \left(\frac{x_1^2 - 1}{x_1} + \frac{4x_1}{x_1^2 - 1} \right) = \frac{(x_1^2 + 1)^2}{4(x_1^3 - x_1)},$$

which is $[2]^*x_1$ by (0.1). Two isogenies with the same pullback of x_1 agree up to sign. Now $c_{\iota \circ \psi \circ \varphi} = 2 \cdot 1 \cdot 1 = 2 = c_{[2]}$ by Problem 1.1(6), while $c_{[-2]} = -2 \neq 2$. So $\iota \circ \psi \circ \varphi = [2]$, and by uniqueness of the dual, $\hat{\varphi} = \iota \circ \psi$, with $c_{\hat{\varphi}} = 2$ and $c_\varphi c_{\hat{\varphi}} = 2 = \deg \varphi$. Finally $\ker \hat{\varphi} = \ker \psi = \{O_2, (0, 0)\} = \varphi(E_1[2](\bar{k}))$ by (1).

(5) Is there an isogeny $\mathbb{G}_m \rightarrow \mathbb{G}_m$ of degree 2? If so, does it have a dual?

SOLUTION. **Yes, $[2]$ and $[-2]$. Their only candidate duals, $[\pm 1]$, have degree 1.** The endomorphisms of $\mathbb{G}_m$ are the maps $[m](t) = t^m$, $m \in \mathbb{Z}$, and $\deg[m] = |m|$, so the isogenies $\mathbb{G}_m \rightarrow \mathbb{G}_m$ of degree 2 are $[2]$ and $[-2]$, both with kernel $\mu_2(\bar{k}) = \{\pm 1\}$. In the sense of the defining identity $\hat{\alpha} \circ \alpha = [\deg \alpha]$, each has a unique dual: $[n] \circ [\pm 2] = [2]$ means $\pm 2n = 2$, so $n = \pm 1$. And in general the only $[n]$ with $[n] \circ [m] = [\deg[m]]$ is $[\text{sign } m]$, which also satisfies

$[m] \circ [\text{sign } m] = [\deg[m]]$ and is compatible with composition. But for $|m| \geq 2$ it fails the other properties of the dual isogeny [Sil09, Theorem III.6.2]: its degree is 1, not $\deg[m]$; the dual of the dual is $[\text{sign } m]$, not $[m]$; and $[m] \mapsto [\text{sign } m]$ is not additive, since $\text{sign}(1+1) \neq \text{sign } 1 + \text{sign } 1$.
Remark. All of this fails because $\deg[m] = |m|$, not m^2 .

3. THE TATE MODULE

We first describe the Tate module of $\mathbb{G}_m$, the familiar case, and then repeat each step for E . What links the two is the Weil pairing, which you may use without proof.

Proposition 3.1 (The Weil pairing [Sil09, Proposition III.8.1]). *Let $m \geq 1$ with $\text{char } k \nmid m$. There is a pairing*

$$e_m: E[m](k^{\text{sep}}) \times E[m](k^{\text{sep}}) \rightarrow \mu_m(k^{\text{sep}})$$

that is bilinear, alternating ($e_m(P, P) = 1$), non-degenerate, and Galois equivariant: $\sigma(e_m(P, Q)) = e_m(\sigma P, \sigma Q)$ for every $\sigma \in \text{Gal}_k$. Here $E[m](k^{\text{sep}}) = E[m](\bar{k})$, since $\text{char } k \nmid m$. It follows that if (P, Q) is a basis of $E[m](k^{\text{sep}})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P, Q)$ is a primitive m -th root of unity, and if σ acts on this basis by the matrix $A \in \text{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma(e_m(P, Q)) = e_m(P, Q)^{\det A}$.

Problem 3.2 (The 2-power torsion of $y^2 = x^3 - x$).

Regard E as a curve over $\mathbb{Q}$, fix $\bar{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\bar{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\bar{\mathbb{Q}})$. Let σ be complex conjugation, acting on $\bar{\mathbb{Q}}$ and on $E(\bar{\mathbb{Q}})$ coordinatewise.

(1) Recall that $T_2(\mathbb{G}_m) := \varprojlim_n \mu_{2^n}(\bar{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$ -basis of $T_2(\mathbb{G}_m)$, and write down its first three terms. How does σ act on $\mu_4(\bar{\mathbb{Q}})$, and on $T_2(\mathbb{G}_m)$?

SOLUTION. $(-1, i, \zeta_8, \dots)$. Also σ is $i \mapsto i^3$ on $\mu_4(\bar{\mathbb{Q}})$ and -1 on $T_2(\mathbb{G}_m)$. The map $\zeta \mapsto \zeta^2$ sends the primitive 2^{n+1} -th roots of unity onto the primitive 2^n -th ones, so there are compatible sequences $(\zeta_{2^n})_n$ of primitive roots with $\zeta_{2^{n+1}}^2 = \zeta_{2^n}$. Since ζ_{2^n} generates the cyclic group $\mu_{2^n}(\bar{\mathbb{Q}})$, the maps $\mathbb{Z}/2^n\mathbb{Z} \rightarrow \mu_{2^n}(\bar{\mathbb{Q}})$, $a \mapsto \zeta_{2^n}^a$, are isomorphisms compatible with reduction on the left and squaring on the right, and in the limit $\mathbb{Z}_2 \xrightarrow{\sim} T_2(\mathbb{G}_m)$. So $(\zeta_{2^n})_n$ is a $\mathbb{Z}_2$ -basis, for instance $(-1, i, \zeta_8, \dots)$ with $\zeta_8 = (1+i)/\sqrt{2} = e^{2\pi i/8}$. Complex conjugation inverts every root of unity: on $\mu_4(\bar{\mathbb{Q}})$ it is $i \mapsto -i = i^3$, and on $T_2(\mathbb{G}_m)$ it is multiplication by $-1 \in \mathbb{Z}_2^\times$.

(2) List $E[2](\bar{\mathbb{Q}})$. Using Problem 1.1(6), find the x -coordinates of all $P \in E(\bar{\mathbb{Q}})$ with $[2]P \in E[2](\bar{\mathbb{Q}}) \setminus \{O\}$, and conclude that $\#E[4](\bar{\mathbb{Q}}) = 16$.

SOLUTION. $E[2](\bar{\mathbb{Q}}) = \{O, (0, 0), (\pm 1, 0)\}$. The x_P are $\pm i, 1 \pm \sqrt{2}, -1 \pm \sqrt{2}$. A point $P \in E(\bar{\mathbb{Q}})$ with $[2]P \in E[2](\bar{\mathbb{Q}}) \setminus \{O\}$ is not in $E[2](\bar{\mathbb{Q}})$, so $4(x_P^3 - x_P) = 4y_P^2 \neq 0$. For such P , evaluating Problem 1.1(6) at P shows that $x([2]P) = ([2]^*x)(P) = e$, that is $[2]P = (e, 0)$, exactly when $g_e(x_P) = 0$: so $x_P = \pm i$ for $e = 0$, $x_P = 1 \pm \sqrt{2}$ for $e = 1$ and $x_P = -1 \pm \sqrt{2}$ for $e = -1$. None of these is a root of $x^3 - x$, so each gives two points $(x_P, \pm y_P) \in E(\bar{\mathbb{Q}})$ outside $E[2](\bar{\mathbb{Q}})$, twelve in all. With $E[2](\bar{\mathbb{Q}})$ this makes 16 points of $E(\bar{\mathbb{Q}})$. They are all of $E[4](\bar{\mathbb{Q}})$, since a point of $E[4](\bar{\mathbb{Q}})$ not in $E[2](\bar{\mathbb{Q}})$ doubles into $E[2](\bar{\mathbb{Q}}) \setminus \{O\}$.

(3) Show that $P_2 := (i, 1-i)$ and $Q_2 := (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\bar{\mathbb{Q}})$, with $[2]P_2 = (0, 0)$ and $[2]Q_2 = (1, 0)$. Show that (P_2, Q_2) is a basis of $E[4](\bar{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\bar{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\bar{\mathbb{Q}})$ and Proposition 3.1.

SOLUTION. For P_2 , $x_{P_2}^3 - x_{P_2} = i^3 - i = -2i = (1-i)^2 = y_{P_2}^2$. For Q_2 , $x_{Q_2}^2 = (1+\sqrt{2})^2 = 3+2\sqrt{2}$, so

$$x_{Q_2}^3 - x_{Q_2} = x_{Q_2}(x_{Q_2}^2 - 1) = 2x_{Q_2}^2 = 6 + 4\sqrt{2} = (2 + \sqrt{2})^2 = y_{Q_2}^2.$$

By (2), $x([2]P_2) = 0$ and $x([2]Q_2) = 1$, and a point of $E(\overline{\mathbb{Q}})$ with x -coordinate 0 or 1 has y -coordinate 0. If $[a]P_2 + [b]Q_2 = O$, doubling gives $[a](0,0) + [b](1,0) = O$, so $a = 2a'$ and $b = 2b'$ are even. Then $[a'](0,0) + [b'](1,0) = O$ makes a', b' even too. So $(\mathbb{Z}/4\mathbb{Z})^2 \rightarrow E[4](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_2 + [b]Q_2$, is injective, hence bijective. Every point of $E[4](\overline{\mathbb{Q}})$ thus lies in $E(\mathbb{Q}(i, \sqrt{2}))$, and $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$, since $\zeta_8 = (1+i)/\sqrt{2}$. This is exactly $\mathbb{Q}(\mu_8(\overline{\mathbb{Q}}))$. Writing $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$ for the field generated by the coordinates of the points of $E[4](\overline{\mathbb{Q}})$, Proposition 3.1 forces $\mathbb{Q}(E[4](\overline{\mathbb{Q}})) \supseteq \mathbb{Q}(\mu_4(\overline{\mathbb{Q}}))$: the values of e_4 lie in $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$ and include a primitive fourth root of unity.

(4) Show that σ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1,0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis (P_2, Q_2) . Compare the determinant of the resulting matrix in $\text{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of σ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Proposition 3.1.

SOLUTION. $\sigma(P_2) = [3]P_2 + [2]Q_2$, $\sigma(Q_2) = Q_2$, determinant $3 \equiv -1$. The group law on $E(\overline{\mathbb{Q}})$ is given by rational functions with coefficients in $\mathbb{Q}$, so $\sigma(P+Q) = \sigma(P) + \sigma(Q)$. In particular $[4]\sigma(P) = \sigma([4]P)$, and σ maps $E[4](\overline{\mathbb{Q}})$ to itself, with inverse σ . Let $R := P_2 + (1,0)$. The line through $(i, 1-i)$ and $(1,0)$ has slope $\lambda = -1$, so

$$x_R = \lambda^2 - i - 1 = -i, \quad y_R = -((1-i) + \lambda(x_R - i)) = -1 - i,$$

that is, $R = (-i, -1-i)$. Hence $\sigma(P_2) = (-i, 1+i) = -R = -P_2 - [2]Q_2$. Since P_2 and Q_2 have order 4, this is $[3]P_2 + [2]Q_2$. And $\sigma(Q_2) = Q_2$, since its coordinates are real. In the basis (P_2, Q_2) , $\sigma|_{E[4](\overline{\mathbb{Q}})}$ has matrix

$$\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \quad \det = 3 \equiv -1 \pmod{4},$$

and by (1), σ acts on $\mu_4(\overline{\mathbb{Q}})$ by the same exponent 3. This is no accident: by Proposition 3.1, $e_4(P_2, Q_2)$ is a primitive fourth root of unity and $\sigma(e_4(P_2, Q_2)) = e_4(P_2, Q_2)^{\det} = e_4(P_2, Q_2)^3$.

(5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0,0)$, $Q_1 = (1,0)$ and P_2, Q_2 as above. Explain why they form a $\mathbb{Z}_2$ -basis of $T_2(E)$, and compare with (1).

SOLUTION. The map $(\mathbb{Z}/2^n\mathbb{Z})^2 \rightarrow E[2^n](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_n + [b]Q_n$, is an isomorphism for every n . It is injective by the argument of (3), by induction on n : if $[a]P_n + [b]Q_n = O$, doubling gives $[a]P_{n-1} + [b]Q_{n-1} = O$, so $a = 2^{n-1}a'$ and $b = 2^{n-1}b'$ by induction. Then $[a']P_1 + [b']Q_1 = O$ makes a', b' even. It is surjective because $\#E[2^n](\overline{\mathbb{Q}}) = 4^n$, as $E[2^n](\overline{\mathbb{Q}}) \cong (\mathbb{Z}/2^n\mathbb{Z})^2$ [Sil09, Corollary III.6.4]. For $n = 2$ this is the count of (2). These isomorphisms are compatible with reduction on the left and $[2]$ on the right. Passing to the limit, $\mathbb{Z}_2^2 \xrightarrow{\sim} T_2(E)$. This is (1) in rank two: $\mu_{2^n}(\overline{\mathbb{Q}})$ is free of rank one over $\mathbb{Z}/2^n\mathbb{Z}$, while $E[2^n](\overline{\mathbb{Q}})$ is free of rank two.

4. MAGMA EXAMPLES

Example 4.1 (The isogenies of Problem 2.2). We check in the function field of E_1 that φ and ψ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with Magma's own 2-isogeny.

```

E := EllipticCurve([-1,0]);           // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;      // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x); y2 := y*(x^2 + 1)/x^2; // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                  // true
x3 := y2^2/x2^2; y3 := y2*(x2^2 - 4)/x2^2; // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                 // true
lam := (3*x^2 - 1)/(2*y); xd := lam^2 - 2*x; yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;            // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));           // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                         // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)

```

Magma's dual is $\iota \circ \psi$ written out in the coordinates x_2, y_2 of E_2 : $((x_2^2 + 4)/(4x_2), y_2(x_2^2 - 4)/(8x_2^2))$.

Example 4.2 (Division polynomials and $E[4](\overline{\mathbb{Q}})$). For $m \geq 2$, $\text{DivisionPolynomial}(E, m)$ vanishes exactly at the x -coordinates of $E[m](\overline{\mathbb{Q}}) \setminus \{O\}$. Its degree is $(m^2 - 1)/2$ for odd m and $(m^2 + 2)/2$ for even m , and its factors for $m = 4$ are $x, x \pm 1$ and the g_e of Problem 1.1(6).

```

[ Degree(DivisionPolynomial(E, m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E, 4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8); i := z^2; s := z + z^7; // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i]; Q2 := E8![1+s, 2+s];
2*P2, 2*Q2; // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4); // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2; // true: complex conjugation

```

Example 4.3 (Characteristic p). Reducing modulo p , the p -division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

```

for p in [3,5,7,13] do
  Ep := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
// 3 true 0 []
// 5 false 10 [ <x^2 + 2, 5> ]
// 7 true 0 []
// 13 false 78 [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]

```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$. In general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod{4}$. For $p = 5, 13$ the division polynomial is the p -th power of a polynomial of degree $(p-1)/2$, whose roots are the x -coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$. The exponent p is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_m$ is $t^p - 1 = (t-1)^p$.

Looking ahead. The Weil pairing of Proposition 3.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map φ_ℓ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]). Formal groups make c_φ an honest derivative. And Frobenius factors $[p]$ in characteristic p .

REFERENCES

- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑